NIGAH

AI VOICE MOBILITY ASSISTANT
FOR THE VISUALLY IMPAIRED

INTELLIGENT VOICE GUIDANCE FOR INDEPENDENT **NAVIGATION.**



EMPOWERING VISUALLY IMPAIRED INDIVIDUALS WITH
VOICE GUIDANCE, SMART DETECTION & TOTAL INDEPENDENCE.

MAJOR PROJECT
2026 – 27

DEPARTMENT OF
COMPUTER SCIENCE & ENGINEERING

TEAM
Nigah Innovators

NiGAH

AI VOICE MOBILITY ASSISTANT

KEY FEATURES

INTELLIGENT NAVIGATION



- Move Left
- Move Right
- Stop
- Step Up
- Step Down

AI VISION



- Object Detection
- Obstacle Detection
- Scene Understanding

VOICE ASSISTANT



- English
- Hindi
- Urdu
- Telugu

SMART READING



- OCR
- Medicine Labels
- Signboards
- Books

FIND MY OBJECT



- Bottle
- Wallet
- Phone
- Keys

SAFETY ALERTS



- Emergency Buzzer
- Voice Alerts

OFFLINE MODE



- Works without Internet

SOS EMERGENCY



- Instant Buzzer
- Voice Alert

DESIGNED TO MAKE NAVIGATION SAFER, SMARTER & INDEPENDENT.

TECHNOLOGY STACK

HARDWARE


Raspberry Pi Zero 2 W


Camera Module


2 x Ultrasonic Sensors


MEMS Microphone


Mini Speaker


Buzzer


Rechargeable Battery


Charging Module

• COMPACT • EFFICIENT • WEARABLE

SOFTWARE

Python

OpenCV

YOLOv8

OCR

Speech-to-Text

Text-to-Speech

</> AI • VISION • VOICE • INTELLIGENCE

PROTOTYPE COST

₹ 2,500

(ESTIMATED COST)

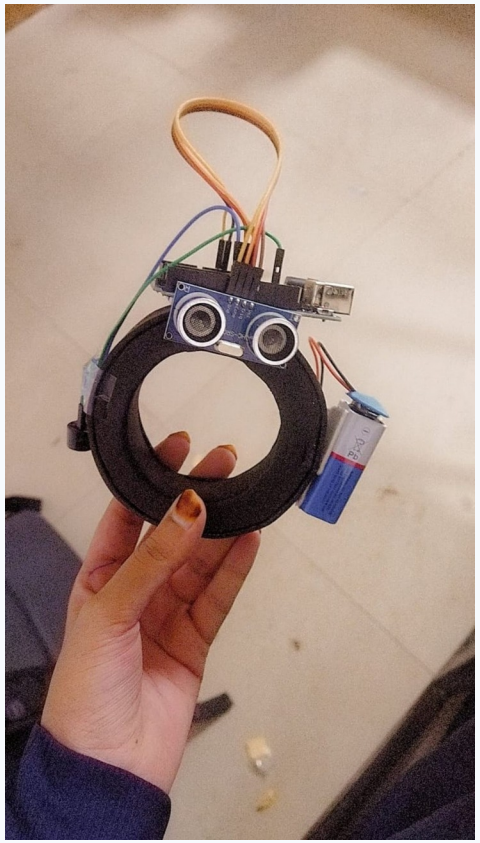
TARGET USERS

- Visually Impaired Individuals
- Schools for the Blind
- NGOs
- Hospitals & Rehabilitation Centers

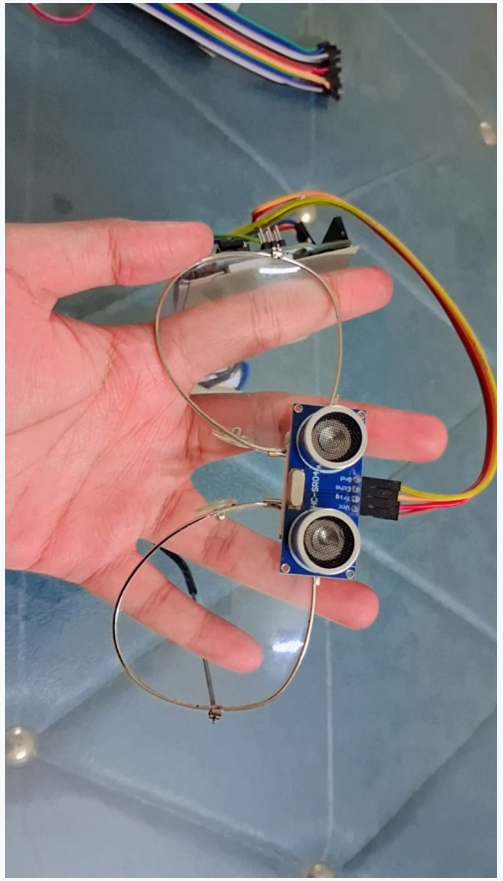
DESIGNED TO MAKE NAVIGATION SAFER, SMARTER & INDEPENDENT.

PROTOTYPE EVOLUTION

From the initial wearable band to the first spectacles prototype



INITIAL PROTOTYPE • WEARABLE BAND



Why we redesigned

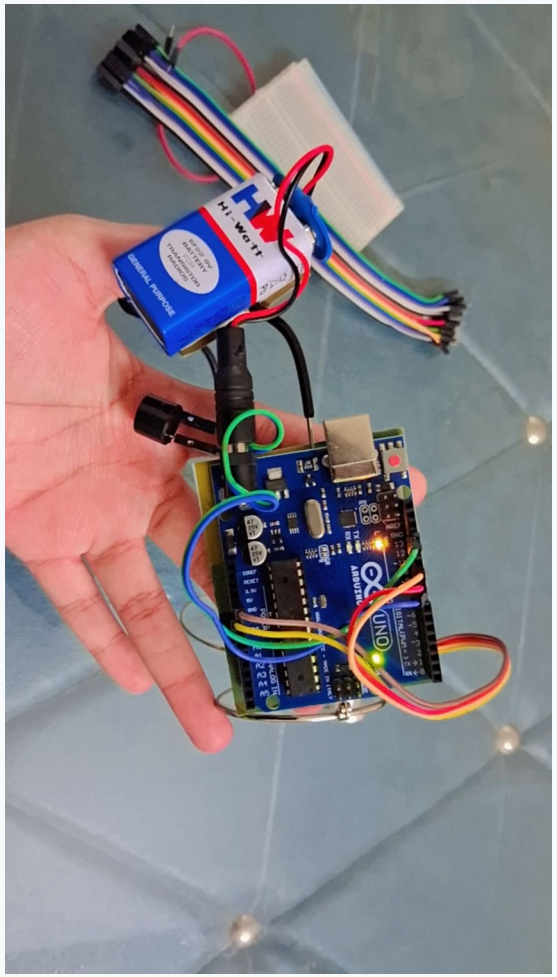
- Better position for head-level sensing.
- Natural, hands-free wearable form.
- Better platform for camera + audio.
- Foundation for the AI layer.

CURRENT

FIRST SPECTACLES PROTOTYPE

CURRENT HARDWARE IMPLEMENTATION

First spectacles prototype • sensor, controller and power integration



Hardware currently integrated

- HC-SR04 ultrasonic sensor — short-range distance sensing
- Arduino UNO — reads sensor data and controls the prototype
- 9V battery / portable supply — independent power
- Temporary wiring for easy testing and tuning

What we are validating

- Sensor placement and orientation on the spectacles.
- Stable distance readings and obstacle response.
- Wearable form-factor feasibility before miniaturization.

HARDWARE VALIDATION → AI INTEGRATION

Why we are validating the sensing architecture before adding the intelligence layer



What we are doing now

- Validate sensor placement and orientation.
- Test distance readings and obstacle response.
- Check power, wiring and physical stability.
- Tune the hardware before adding more complexity.

Why AI comes next

- AI should be tested on a stable hardware foundation.
- This separates hardware issues from AI issues.
- Next prototype: camera + computer vision + voice guidance.
- Then integrate the three advanced intelligence features.

CURRENT = HARDWARE FOUNDATION → NEXT = AI INTELLIGENCE

NIGAH SYSTEM FLOW

How the final system will sense, understand and guide

01

ENVIRONMENT

Camera +
Ultrasonic Sensors

02

PERCEPTION

Object / Hazard
Detection

03

AI DECISION

Understand +
Prioritize

04

GUIDANCE

Voice Instruction
+ Buzzer

05

USER ACTION

Move / Stop /
Avoid

Example

- Sensor / camera: obstacle ahead
- AI: decides it is relevant
- Guidance: “Pole ahead. Stop.”

Core idea

- Detect → Understand → Decide → Guide
- The system should provide an action, not only an object name.

INTELLIGENCE ROADMAP

Three selected innovations for the next development phase

01 AI VISUAL MEMORY

Remembers where important objects were last seen.

“Where are my glasses?”
→ “Last seen on the study table.”

02 ENVIRONMENTAL HAZARD INTELLIGENCE

Identifies hazards beyond ordinary objects.

Wet floor • Construction area • Open manhole

03 ADAPTIVE ROUTE LEARNING

Learns frequently travelled routes and provides more personalized guidance.

Repeated route
→ Learned path
→ Smarter guidance

NIGAH evolves from sensing the present → remembering the past → learning the user’s routes.